

# SIMATS ENGINEERING

## ASSESSMENT TOOL3:REVERSEENGINEERING TASK

### Cloud ComputingandBigDataAnalytics–CSA1518

**CO4: Analyze big data characteristics and challenges across domains including security and application aspects.**

Question Count: 5 | Total Marks: 30

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### Question 1:Website andCustomerTransactionBigData Platform

The platform collects billions of website clicks, mobile interactions, and customer transactions every day. Because the data volume is extremely large and comes from structured and semi-structured sources, a distributed Big Data architecture is likely required.

#### 1. Understanding of the Given System

A probable architecture is:

#### Probable Architecture

| Web / Mobile / Transactions | Data Ingestion | Distributed Storage | Batch+Stream Processing | Analytics | BI Dashboards |
|-----------------------------|----------------|---------------------|-------------------------|-----------|---------------|
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#### 2. Probable Distributed Storage System

A distributed storage system such as HDFS or cloud object storage is likely to be used. Structured transaction data can be stored in structured tables or a data warehouse, while semi-structured clickstream and mobile- event data can be stored in a data lake. Replication would probably protect data from node failures, and partitioning could be based on date, region, or event type.

#### 3. Probable Data IngestionFramework

A distributed event-ingestion framework such as Apache Kafka is a likely choice for continuously receiving website clicks, mobile interactions, and customer transactions. Kafka partitions can support parallel ingestion.

#### 4. Probable Batch andStreamProcessing

Apache Spark is likely for large-scale batch analytics, while Spark Structured Streaming or a similar streaming engine is likely for near-real-time analytics.

#### 5. Structured and Semi-Structured Data

| Data                              | Probable Storage                     |
|-----------------------------------|--------------------------------------|
| Customer transactions             | Structured database / data warehouse |
| Website clicks                    | Data lake / semi-structured storage  |
| Mobile events Analytical datasets | Data warehouse / analytical store    |

## 6. Scalability and Partitioning

- Horizontal scaling by adding storage and processing nodes.
- Kafka partitions for parallel event ingestion.
- Data partitioning by date, region, or event type.
- Replication for fault tolerance.
- Distributed processing to divide large workloads among machines.

## 7. Dashboard and BI Integration

The analytics layer would likely feed a data warehouse or analytical database, which can then connect to Business Intelligence dashboards. Dashboards could display customer activity, transaction trends, and business KPIs.

## 8. Reverse-Engineering Conclusion

The massive data volume and continuous event generation strongly suggest a distributed architecture. Kafka for ingestion, HDFS/cloud object storage for distributed storage, Spark for batch and streaming analytics, and a warehouse/BI layer are probable components inferred from the requirements.

## Question 2: E-Commerce Personalization Platform

The e-commerce platform tracks cart additions, abandoned checkouts, product views, and payment behavior across regions. It requires real-time personalization, customer profiling, recommendations, and KPI monitoring.

### Probable E-Commerce Architecture

| User Events | Event Streaming | Stream Processing | Customer Profiles / Features | Recommendation ML | Personalized Output / KPIs |
|-------------|-----------------|-------------------|------------------------------|-------------------|----------------------------|
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### 1. Probable Event Streaming Tools

Apache Kafka is a likely event-streaming platform. Separate topics could be used for product views, cart additions, checkout events, and payment events. Partitions would allow parallel processing across regions.

### 2. Customer Profiling Database

A NoSQL database is likely for rapidly changing customer profiles containing viewed products, cart history, purchase history, preferences, and recent activity. Structured order and payment records could be maintained separately.

### 3. Real-Time Synchronization

Session data, clickstream logs, and transaction records can be synchronized through the streaming layer:

Click/Product View → Kafka → Stream Processing → Customer Profile Update → Recommendation Model → Personalized Recommendation

### 4. Probable Caching and Query Systems

A fast in-memory cache such as Redis is likely for active sessions, frequently accessed products, recent preferences, and recommendations. A distributed query engine such as Presto/Trino could query large datasets across distributed storage.

## 5. Machine Learning Workflow

A probable recommendation workflow is: Historical Data → Feature Engineering → Model Training → Recommendation Model → Real-Time Customer Features → Recommendation. Collaborative filtering, content-based, or hybrid approaches are possible.

## 6. Structured and Semi-Structured Data

| Data                          | Examples                                                 |
|-------------------------------|----------------------------------------------------------|
| Structured order data         | Order ID, customer ID, product ID, price, payment status |
| Semi-structured browsing data | Clicks, product views, searches, session events          |

A distributed processing layer can join these datasets using identifiers such as customer ID and product ID.

## 7. Scalability and Fault Tolerance

- Kafka partitions for parallel event processing.
- Replication for fault tolerance.
- Distributed databases for large customer profiles.
- Multiple processing nodes.
- Caching to reduce database load.
- Regional processing to reduce latency.

## 8. Dashboard and KPI Monitoring

Processed data can feed a dashboard showing conversion rate, cart abandonment, product views, purchase rate, recommendation engagement, and regional performance.

## 9. Reverse-Engineering Conclusion

The real-time personalization requirement strongly suggests event streaming, distributed processing, customer-profile storage, caching, and machine learning. Kafka, Spark, NoSQL, Redis, and a distributed query engine are probable components.

## Question 3: Healthcare Big Data System

The healthcare system processes patient histories, diagnostic images, wearable sensor streams, and prescription records. It connects hospitals, laboratories, and insurance systems and supports disease prediction, monitoring, clinical decision-making, and population health studies.

### Probable Healthcare Big Data Architecture

| Hospitals / Labs / Wearables | Integration / ETL | Hybrid Storage | Stream + Batch Processing | ML / Analytics | Clinical / Population Insights |
|------------------------------|-------------------|----------------|---------------------------|----------------|--------------------------------|
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## 1. Probable Hybrid Database Architecture

| Data Type                   | Probable Storage             |
|-----------------------------|------------------------------|
| Patient records             | Relational database          |
| Prescription records        | Relational database          |
| Diagnostic images           | Object / distributed storage |
| Wearable Analytics datasets | Time-series / NoSQL storage  |
|                             | Data lake / data warehouse   |

The hybrid design is inferred from the different structures and access patterns of the healthcare data.

## 2. Probable Data Integration Tools

An integration/ETL layer is likely to extract data from hospitals, laboratories, and insurance systems, transform it into common formats, validate it, remove duplicates, and load it into the analytics platform. HL7/FHIR are probable candidates for healthcare information exchange, although the case does not mandate a specific standard.

## 3. Streaming and Machine Learning

Wearable devices generate continuous streams. A probable flow is Wearables → Stream Ingestion → Stream Processing → Monitoring. Machine-learning frameworks can use historical and current data for disease-risk prediction and monitoring.

## 4. Privacy, Encryption and Compliance

Encryption during transmission and storage.  
Strong authentication and role-based access control. Audit logs for sensitive-data access.  
Data minimization and controlled sharing.  
Monitoring and governance mechanisms.

The case requires privacy, encryption, and compliance, but it does not name a specific regulation; therefore no particular regulatory framework is assumed here.

## 5. Analytics Pipeline

Patient Data → Data Integration → Quality Validation → Storage → Distributed Processing → ML/Analytics → Clinical Decision Support

For population studies: Historical Patient Data → Distributed Processing → Statistical/ML Analysis → Population Health Insights.

## 6. Reverse-Engineering Conclusion

The combination of structured records, unstructured images, streaming wearable data, and multi-organization integration strongly indicates hybrid storage, an integration layer, streaming/batch processing, ML analytics, and strong privacy controls.

## Question 4: Smart City Big Data Platform

The smart-city platform continuously receives traffic sensor feeds, CCTV metadata, weather updates, and public transport activity. It requires real-time congestion prediction and batch analytics for urban planning.

**Probable Smart City Architecture**

| Sensors / CCTV / Weather / Transport | Edge Computing | Distributed Messaging | Stream + Batch Analytics | Time-Series / Geospatial Storage | Live Operations Dashboard |
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## 1. Probable Edge Computing Setup

Edge computing is likely near traffic sensors and other city devices to collect data, perform basic filtering, detect immediate events, and reduce the amount of data transmitted to centralized systems.

## 2. Probable Distributed Messaging

Apache Kafka is a probable messaging framework. Traffic, weather, public transport, and CCTV metadata can be separated into topics, with partitions supporting parallel consumption.

## 3. Centralized and Specialized Storage

| Data / Requirement                | Probable Technology                          |
|-----------------------------------|----------------------------------------------|
| Large historical datasets         | Distributed data lake / cloud object storage |
| Sensor measurements over time     | Time-series database                         |
| Location-based queries Historical | Geospatial database                          |
| analytical reporting              | Data warehouse                               |

## 4. Geospatial and Streaming Analytics

Streaming analytics can identify congestion, sudden traffic changes, public transport delays, and weather-related effects. Geospatial processing can associate events with roads, intersections, routes, and geographic regions.

## 5. Batch and Real-Time Analytics

Real-time: Traffic Stream → Stream Processing → Congestion Detection → Live Operations Dashboard

Batch: Historical Traffic + Weather + Transport Data → Distributed Processing → Urban Planning Analytics

## 6. Security and Access Control

- Authentication and role-based access control.
- Encryption of sensitive data.
- Network security controls.
- Access logging and monitoring.
- Separation of public, operational, and restricted information.

## 7. Reverse-Engineering Conclusion

Continuous sensor feeds strongly suggest edge computing, distributed messaging, stream processing, and specialized time-series and geospatial storage. Combining real-time and batch analytics supports both immediate city operations and long-term planning.

## Question 5: Healthcare Analytics System

The healthcare analytics system integrates patient records, wearable readings, laboratory reports, and appointment histories. It requires hybrid storage, interoperability, real-time monitoring, predictive analytics, distributed processing, and strong data governance.

### Probable Healthcare Analytics Architecture

| Hospitals / Clinics / Labs / Wearables | ETL + Interoperability | Hybrid Storage | Stream + Distributed Batch Processing | ML Analytics | Risk Scoring / Recommendations |
|----------------------------------------|------------------------|----------------|---------------------------------------|--------------|--------------------------------|
| ↓                                      | ↓                      | ↓              | ↓                                     | ↓            |                                |

## 1. Probable Hybrid Storage Architecture

| Data                | Probable Storage                     |
|---------------------|--------------------------------------|
| Patient records     | Relational database                  |
| Appointment history | Relational database                  |
| Laboratory reports  | Structured / semi-structured storage |

|                                            |                                                   |
|--------------------------------------------|---------------------------------------------------|
| Wearable readings Large analytics datasets | Time-series / NoSQL storage Distributed data lake |
|--------------------------------------------|---------------------------------------------------|

## 2. Probable ETL Pipeline

Extract → Validate → Transform → Standardize → Integrate → Load

The pipeline can extract data from hospitals, clinics, laboratories, and devices; standardize formats; validate quality; remove duplicates; integrate records; and load the resulting datasets for analysis.

## 3. Interoperability

Standardized healthcare data exchange is likely required. HL7/FHIR are probable interoperability standards, although the assessment does not explicitly mandate a particular standard.

## 4. Real-Time Monitoring

Wearable → Stream Ingestion → Stream Processing → Patient Monitoring → Alert

Continuous wearable readings can be processed to identify abnormal measurements and support timely monitoring.

## 5. Predictive Health Analytics

Historical Data → Feature Engineering → ML Model → Risk Score → Clinical Support

Classification, regression, or anomaly-detection approaches may support risk prediction. The exact model cannot be determined from the case alone.

## 6. Distributed Processing

A distributed framework such as Apache Spark is likely for population-level analysis. Large patient datasets

can be divided across multiple processing nodes to improve scalability and reduce processing time.

## 7. Encryption, Compliance and Auditing

- Encryption at rest and in transit.
- Authentication and role-based authorization.
- Audit logs and access monitoring.
- Data retention and governance policies.
- Controlled access to sensitive medical information.

The case requires compliance but does not specify a particular regulation; therefore no specific regulatory framework is claimed as mandatory.

## 8. Machine Learning Support

ML may support diagnosis-risk scoring, early warning, patient risk stratification, and treatment-support recommendations. These outputs should be treated as analytical support rather than autonomous medical decisions.

## 9. Reverse-Engineering Conclusion

The mixture of structured medical records, wearable streams, laboratory information, and appointment history strongly indicates hybrid storage with ETL/interoperability, real-time streaming, distributed processing, and ML analytics. Encryption, access control, and auditing are likely because the system handles sensitive medical information.